

Decent Coaching Classes

VIII CBSE

Worksheet

Mathematics | Chapter 6: We Distribute, Yet Things Multiply

SECTION A — Multiple Choice Questions

- Multiply $3x$ by $(2x^2 - 5)$.
(a) $6x^3 - 5$ (b) $6x^3 - 15x$ (c) $5x^3 - 15x$ (d) $6x^2 - 15x$
- $(x + 3)(x + 4) =$
(a) $x^2 + 12$ (b) $x^2 + 7x + 12$ (c) $x^2 + 7x + 7$ (d) $x^2 + 12x + 7$
- $(a + b)^2 =$
(a) $a^2 + b^2$ (b) $a^2 + 2ab + b^2$ (c) $a^2 - 2ab + b^2$ (d) $a^2 + ab + b^2$
- $(2x - 3)^2 =$
(a) $4x^2 - 9$ (b) $4x^2 + 12x + 9$ (c) $4x^2 - 12x + 9$ (d) $2x^2 - 12x + 9$
- The value of 102^2 is
(a) 10004 (b) 10404 (c) 10204 (d) 10424
- Using an identity, $96 \times 104 =$
(a) 9984 (b) 9996 (c) 10016 (d) 9904
- $57 \times 11 =$
(a) 617 (b) 627 (c) 567 (d) 657
- $43 \times 101 =$
(a) 4343 (b) 4403 (c) 4301 (d) 4433
- $(x + 1/x)^2 =$
(a) $x^2 + 1/x^2$ (b) $x^2 + 2 + 1/x^2$ (c) $x^2 - 2 + 1/x^2$ (d) $x^2 + 1 + 1/x^2$
- $x^2 - 9$ can be written as
(a) $(x - 3)^2$ (b) $(x + 3)^2$ (c) $(x - 3)(x + 3)$ (d) $(x - 9)(x + 1)$
- $2a(a + b) - a(2a - b)$ simplifies to
(a) ab (b) $2ab$ (c) $3ab$ (d) $4a^2 + ab$
- $(x + 2)(x - 2)(x^2 + 4) =$
(a) $x^4 + 16$ (b) $x^4 - 16$ (c) $x^4 - 8$ (d) $x^2 - 16$
- $47 \times 99 =$
(a) 4653 (b) 4753 (c) 4600 (d) 4663
- If $x + y = 6$ and $xy = 8$, then $x^2 + y^2 =$
(a) 20 (b) 28 (c) 36 (d) 44
- The number of terms in the simplified product $(x + 2)(x + 3)$ is
(a) 2 (b) 3 (c) 4 (d) 5
- $(3x + 2y)(3x - 2y) =$
(a) $9x^2 + 4y^2$ (b) $9x^2 - 4y^2$ (c) $6x^2 - 4y^2$ (d) $9x^2 - 2y^2$

SECTION B — Fill in the Blanks

- $(a - b)^2 = a^2 - \underline{\hspace{2cm}} + b^2$.
- $(x + a)(x + b) = x^2 + (a + b)x + \underline{\hspace{2cm}}$.
- $(a + b)(a - b) = \underline{\hspace{2cm}}$.
- $25^2 = (20 + 5)^2 = \underline{\hspace{2cm}}$.
- $63 \times 11 = \underline{\hspace{2cm}}$.
- $76 \times 99 = 7600 - 76 = \underline{\hspace{2cm}}$.
- $(x + 2)(x + 2) = \underline{\hspace{2cm}}$.
- The value of $10y^0$ ($y \neq 0$) is $\underline{\hspace{2cm}}$.

SECTION C — True or False

25. $(a + b)^2 = a^2 + b^2$.
26. $(x - y)^2 = (y - x)^2$.
27. $101 \times 37 = 3737$.
28. $(x + 3)(x - 3) = x^2 + 9$.
29. The product of a monomial and a binomial has two terms.
30. $99^2 = 9801$.

SECTION D — Short Answer Questions

Multiplying Algebraic Expressions

31. Expand using the distributive property: $(\frac{3}{4})(a + b - \frac{2}{3})$.
32. Multiply $5ab$ by $(-2a^2 + 3ab)$.
33. Simplify: $4x(5x^2 + 3x)$.
34. Multiply $(13a^2b - 7abc)$ by $4abc$.
35. Find the product: $(7ab + 5)(6ab + 3)$.
36. Find the product: $(x - 2)(x^2 + x + 1)$.
37. Find the value of $2x^2 \times (-3xy) \times y^2$ when $x = 1$ and $y = 2$.
38. Simplify: $(x + 2)(3x + 4) - 3x(x + 2)$.
39. Simplify: $(a + b)(a + 2b) - (a - b)(a - 2b)$.
40. A student simplified $3(y - 2) + 4(y + 5)$ as $3y - 2 + 4y + 5 = 7y + 3$. Is there a mistake? If yes, write the correct expression.

Multiplication Tricks Using the Distributive Property

41. Write the product in one line: (i) 78×11 (ii) 435×11 (iii) 356×101
42. Use the distributive property to find: (i) 94×101 (ii) 743×1001 (iii) 6725×99 (iv) 3487×999

Identities

43. Expand: (i) $(7y + 6)^2$ (ii) $(c - 7)^2$ (iii) $(-3x + 5)^2$
44. Evaluate using identities: (i) 88^2 (ii) 213^2 (iii) 86×94 (iv) 104×105
45. Expand: (i) $(x - y^2)^2$ (ii) $(x^2/4 + y^2/2)^2$ (iii) $(3x^2 + 1/x)(3x^2 - 1/x)$
46. Find the product: (i) $(xy + z)(xy + z)$ (ii) $(6x - 9y)(6x - 9y)$ (iii) $(5yz + 6x^2)(5yz - 6x^2)$ (iv) $(2a^3 - b^3)(2a^3 + b^3)$
47. Simplify: (i) $(3x + 2)^2 - (3x - 2)^2$ (ii) $(ab + bc)^2 - 2ab^2c$
48. Using an identity, evaluate: (i) $96^2 - 4^2$ (ii) $61^2 - 39^2$ (iii) $215^2 - 205^2$ (iv) $850^2 - 750^2$
49. Using the identities, evaluate: (i) $(94 \times 94 - 6 \times 6)/(94 - 6)$ (ii) $(112 \times 112 - 50 \times 50)/(112 + 50)$
50. Find the value of x : (i) $12x = 50 \times 50 - 38 \times 38$ (ii) $60x = 53^2 - 37^2$ (iii) $24x = 72^2 - 48^2$
51. Find the continued products: (i) $(x + 3)(x - 3)(x^2 + 9)$ (ii) $(2x - 1)(2x + 1)(4x^2 + 1)$

SECTION E — Long Answer and Application Questions

52. A three-digit number is written as $100a + 10b + c$. Show algebraically that multiplying it by 1001 repeats its digits, and hence find 743×1001 .
53. Find the product $(3x + 4y)(3x - 4y)(9x^2 + 16y^2)$ and verify the result for $x = 1$ and $y = 2$.
54. If $a + b = 9$ and $ab = 20$, find (i) $a^2 + b^2$ (ii) $(a - b)^2$.
55. If $x - 1/x = 4$, find (i) $x^2 + 1/x^2$ (ii) $(x + 1/x)^2$.
56. A rectangular garden has length $(3x + 2)$ m and breadth $(2x - 1)$ m. Write an expression for its area and find the area when $x = 5$.
57. A square plot has side $(2a + 3b)$ m. Find its area as an expression, and its value when $a = 3$ and $b = 1$.
58. Using identities, evaluate (i) 998×1002 (ii) 1005^2 .
59. Prove that $(a + b)^2 - (a - b)^2 = 4ab$. Hence evaluate $(25 + 15)^2 - (25 - 15)^2$.

ANSWER KEY

Section A — Multiple Choice Questions

1. (b)	2. (b)	3. (b)	4. (c)	5. (b)	6. (a)	7. (b)	8. (a)
9. (b)	10. (c)	11. (c)	12. (b)	13. (a)	14. (a)	15. (b)	16. (b)

Section B — Fill in the Blanks

17. $2ab$
18. ab
19. $a^2 - b^2$
20. $400 + 200 + 25 = 625$
21. 693
22. 7524
23. $x^2 + 4x + 4$
24. 10

Section C — True or False

25. False; $(a + b)^2 = a^2 + 2ab + b^2$
26. True
27. True
28. False; $(x + 3)(x - 3) = x^2 - 9$
29. True
30. True; $(100 - 1)^2 = 10000 - 200 + 1$

Section D — Short Answer Questions

31. $3a/4 + 3b/4 - 1/2$.
32. $-10a^3b + 15a^2b^2$.
33. $20x^3 + 12x^2$.
34. $52a^3b^2c - 28a^2b^2c^2$.
35. $42a^2b^2 + 21ab + 30ab + 15 = 42a^2b^2 + 51ab + 15$.
36. $x^3 + x^2 + x - 2x^2 - 2x - 2 = x^3 - x^2 - x - 2$.
37. Product = $-6x^3y^3 = -6(1)(8) = -48$.
38. $3x^2 + 10x + 8 - 3x^2 - 6x = 4x + 8$.
39. $(a^2 + 3ab + 2b^2) - (a^2 - 3ab + 2b^2) = 6ab$.
40. Yes. 3 was not multiplied with -2 , and 4 was not multiplied with 5. Correct: $3y - 6 + 4y + 20 = 7y + 14$.
41. (i) 858 (ii) 4785 (iii) 35956
42. (i) $9400 + 94 = 9494$ (ii) $743000 + 743 = 743743$ (iii) $672500 - 6725 = 665775$ (iv) $3487000 - 3487 = 3483513$
43. (i) $49y^2 + 84y + 36$ (ii) $c^2 - 14c + 49$ (iii) $9x^2 - 30x + 25$
44. (i) $(90 - 2)^2 = 7744$ (ii) $(200 + 13)^2 = 45369$ (iii) $(90 - 4)(90 + 4) = 8084$ (iv) $(100 + 4)(100 + 5) = 10920$
45. (i) $x^2 - 2xy^2 + y^4$ (ii) $x^4/16 + x^2y^2/4 + y^4/4$ (iii) $9x^4 - 1/x^2$
46. (i) $x^2y^2 + 2xyz + z^2$ (ii) $36x^2 - 108xy + 81y^2$ (iii) $25y^2z^2 - 36x^4$ (iv) $4a^6 - b^6$
47. (i) $24x$ (ii) $a^2b^2 + 2ab^2c + b^2c^2 - 2ab^2c = a^2b^2 + b^2c^2$
48. (i) $92 \times 100 = 9200$ (ii) $22 \times 100 = 2200$ (iii) $10 \times 420 = 4200$ (iv) $100 \times 1600 = 160000$
49. (i) $(94 + 6) = 100$ (ii) $(112 - 50) = 62$
50. (i) $12x = 12 \times 88$, so $x = 88$ (ii) $60x = 16 \times 90 = 1440$, so $x = 24$ (iii) $24x = 24 \times 120$, so $x = 120$
51. (i) $(x^2 - 9)(x^2 + 9) = x^4 - 81$ (ii) $(4x^2 - 1)(4x^2 + 1) = 16x^4 - 1$

Section E — Long Answer and Application Questions

52. $(100a + 10b + c)(1000 + 1) = 100000a + 10000b + 1000c + 100a + 10b + c$.
This is the six-digit number $abcabc$, so the digits repeat.
 $743 \times 1001 = 743743$.
53. $(9x^2 - 16y^2)(9x^2 + 16y^2) = 81x^4 - 256y^4$.
LHS: $(3 + 8)(3 - 8)(9 + 64) = 11 \times (-5) \times 73 = -4015$.
RHS: $81 - 256 \times 16 = 81 - 4096 = -4015$. Verified.

54. (i) $a^2 + b^2 = (a + b)^2 - 2ab = 81 - 40 = 41$.
 (ii) $(a - b)^2 = (a + b)^2 - 4ab = 81 - 80 = 1$.
55. (i) $x^2 + 1/x^2 = (x - 1/x)^2 + 2 = 16 + 2 = 18$.
 (ii) $(x + 1/x)^2 = x^2 + 1/x^2 + 2 = 18 + 2 = 20$.
56. Area = $(3x + 2)(2x - 1) = 6x^2 - 3x + 4x - 2 = 6x^2 + x - 2$.
 For $x = 5$: $150 + 5 - 2 = 153 \text{ m}^2$. (Check: $17 \times 9 = 153$.)
57. Area = $(2a + 3b)^2 = 4a^2 + 12ab + 9b^2$.
 For $a = 3$, $b = 1$: $36 + 36 + 9 = 81 \text{ m}^2$. (Check: side = 9 m.)
58. (i) $(1000 - 2)(1000 + 2) = 1000000 - 4 = 999996$.
 (ii) $(1000 + 5)^2 = 1000000 + 10000 + 25 = 1010025$.
59. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.
 Hence $(25 + 15)^2 - (25 - 15)^2 = 4 \times 25 \times 15 = 1500$.